

Radial-velocity benchmark report:

LL Aquarii and AK Fornacis

Broadening-function velocities from RVCalPy (<https://github.com/atemelci/RVCalPy>) compared with published measurements

2026 August 10

Velocities measured with RVCalPy reproduce the published HARPS velocities of LL Aqr to 0.092 and 0.142 km s⁻¹ (primary, secondary) after removal of a constant zero-point offset, with reduced χ^2 near unity — the differences are fully explained by the quoted uncertainties. For AK For the primary behaves the same way (0.150 km s⁻¹), while the secondary shows a systematic velocity-amplitude deficit of 0.8% traced to the temperature mismatch of the single template. This report states what was compared, how the statistics are defined, and what the result means for masses derived from these velocities.

1. What was tested

Radial velocities measured with RVCalPy were compared, night by night, with the published velocities of two double-lined eclipsing binaries. Only epochs present in both data sets were used, so nothing in this report depends on data that cannot be checked against a published number: 16 nights for LL Aqr and 14 for AK For. Our spectra are public ESO archive phase-3 products; the measurement is a singular-value-decomposition broadening function against a synthetic template over 500–550 nm.

	LL Aqr	AK For
Comparison paper	Graczyk et al. (2016)	Helminiak et al. (2014)
Instrument(s)	HARPS	FEROS (14) + HARPS (2)
Spectra measured / compared	17 / 16	16 / 14
Template (Teff)	HIP 57757 (6093 K)	HIP 108870 (4676 K)
BF step Δv	1 km s ⁻¹	2 km s ⁻¹
Orbital period	20.1784 d	3.98099 d
Epochs excluded	1 (not in their table)	2 ours + 6 CORALIE rows

Table 1. Material and settings.

2. What we compared against

	LL Aqr	AK For
Authors	Graczyk, D., Smolec, R., Gazeas, K., et al.	Helminiak, K. G., Graczyk, D., Konacki, M., et al.
Title	A solar twin in the eclipsing binary LL Aqr	Orbital and physical parameters of eclipsing binaries from the ASAS catalogue — VI. AK Fornacis: a rare, bright K-type eclipsing binary
Journal	A&A 594, A92 (2016)	A&A 567, A64 (2014)
DOI	10.1051/0004-6361/201628918	10.1051/0004-6361/201220985
Preprint	arXiv:1608.01000	arXiv:1405.2887
Table used	Table 1 — individual RVs (HARPS rows)	Table 2 — individual RVs, errors and residuals
Rows used	16 of 32 (all HARPS rows; the CORALIE rows have no counterpart here)	14 of 20 (12 FEROS + 2 HARPS; the 6 CORALIE rows have no counterpart here)

Table 2. The published measurements used as the reference.

Both tables were transcribed from the LaTeX source of the preprints rather than from rendered PDFs, and checked row by row. Method: Rucinski (1992, AJ 104, 1968; 2002, AJ 124, 1746). Pipeline: RVCalPy, <https://github.com/atemelci/RVCalPy>.

3. How the numbers are defined

For each matched night we take the difference between our velocity and the published one, with the two uncertainties added in quadrature:

$$d_i = v_i^{\text{ours}} - v_i^{\text{pub}}, \quad \sigma_i = \sqrt{\varepsilon_i^{2, \text{ours}} + \varepsilon_i^{2, \text{pub}}}, \quad w_i = \sigma_i^{-2}$$

Δ is the weighted mean offset (a difference in velocity zero point, absorbed by the systemic velocity in any orbit fit and therefore harmless), s the scatter about it, χ_r^2 the check of whether that scatter is explained by the quoted uncertainties, and p the residual expressed in units of its own uncertainty (plotted in the lower panels of the figures):

$$\Delta = \frac{\sum w_i d_i}{\sum w_i}, \quad s = \sqrt{\frac{\sum (d_i - \Delta)^2}{N - 1}}, \quad \chi_r^2 = \frac{1}{N - 1} \sum \left(\frac{d_i - \Delta}{\sigma_i} \right)^2, \quad p_i = \frac{d_i - \Delta}{\sigma_i}$$

A slope a from the straight-line fit $v(\text{ours}) = a \cdot v(\text{published}) + b$ separates the two failure modes: $a = 1$ with $b \neq 0$ is a pure zero point, while $a \neq 1$ is an amplitude error that propagates into the masses.

4. Results

	LL Aqr		AK For	
	Primary	Secondary	Primary	Secondary
Nights compared	16	16	14	14
Offset Δ (km s ⁻¹)	+0.251	+0.294	+0.208	+0.318
$\varepsilon(\Delta)$ (km s ⁻¹)	0.023	0.040	0.059	0.130
Scatter s (km s ⁻¹)	0.092	0.142	0.150	0.864
Reduced χ^2	1.03	0.81	0.45	2.54
rms of pulls	0.98	0.87	0.65	1.54
Slope a	0.9997	0.9989	0.9986	0.9923
Amplitude error	-0.03%	-0.11%	-0.14%	-0.77%
Their own fit rms	0.053	0.049	0.082	0.096

Table 3. Summary of both comparisons. The last row is the residual rms of the authors' own orbital fits, for scale.

5. Verdict

- **LL Aqr.** After removing offsets of +0.251 and +0.294 km s⁻¹, the residuals scatter by only 0.092 and 0.142 km s⁻¹ with reduced χ^2 of 1.03 and 0.81. Values near unity mean the differences are fully explained by the quoted uncertainties: there is no unexplained disagreement. Velocity amplitudes are reproduced to better than 0.15%, so an orbit fitted to our velocities returns the same masses.
- **AK For.** The primary behaves like LL Aqr (0.150 km s⁻¹, $\chi_r^2 = 0.45$). The secondary does not: $\chi_r^2 = 2.54$, and the residuals run from -1.5 km s⁻¹ near phase 0.1 to +1.3 km s⁻¹ beyond phase 0.6 — a systematic pattern, not noise. The line fit identifies it as an amplitude deficit of 0.8% rather than a zero-point shift.
- **Cause and consequence.** The single 4676 K template matches the AK For primary (4690 ± 100 K) but is ~300 K hotter than the secondary (4390 ± 150 K); both stars are also spotted and active, and our 50 nm window contains fewer clean secondary lines than the full range used in the published analyses. Because K_2 sets the mass of the primary, an orbit fitted to our AK For velocities alone would underestimate M_1 by about one per cent. Re-measuring the secondary against a ~4400 K template is the obvious test. The four zero-point offsets themselves (all +0.2 to +0.3 km s⁻¹) have the same origin and matter only if the absolute velocity is itself the measurement.

Appendix A. Night-by-night residuals

Phase	Δv_1	σ_1	Δv_2	σ_2	p_1	p_2
0.0689	+0.237	0.093	+0.241	0.164	-0.16	-0.33
0.0921	+0.234	0.096	+0.253	0.167	-0.18	-0.25
0.1182	+0.329	0.093	+0.185	0.166	+0.84	-0.66
0.1243	+0.274	0.093	+0.248	0.165	+0.24	-0.28
0.1302	+0.233	0.094	+0.261	0.165	-0.20	-0.20
0.1560	+0.363	0.093	+0.131	0.166	+1.19	-0.98
0.2781	+0.316	0.093	+0.149	0.158	+0.70	-0.92
0.3023	+0.074	0.091	+0.411	0.161	-1.94	+0.73
0.4344	+0.058	0.089	+0.656	0.152	-2.17	+2.38
0.4833	+0.419	0.091	+0.202	0.157	+1.84	-0.59
0.6434	+0.290	0.093	+0.124	0.159	+0.42	-1.07
0.6555	+0.290	0.092	+0.236	0.159	+0.43	-0.37
0.6927	+0.224	0.093	+0.408	0.163	-0.30	+0.70
0.7058	+0.201	0.097	+0.392	0.163	-0.51	+0.60
0.7258	+0.226	0.095	+0.452	0.164	-0.27	+0.96
0.8864	+0.261	0.091	+0.318	0.167	+0.10	+0.14

Table A.1. LL Aqr (HARPS). Differences (this work – published) in km s^{-1} , their combined uncertainties, and the pulls $p = (d - \Delta)/\sigma$.

Phase	Inst.	Δv_1	σ_1	Δv_2	σ_2	p_1	p_2
0.0979	FEROS	+0.348	0.229	-1.509	0.551	+0.61	-3.31
0.1380	FEROS	+0.320	0.271	-1.644	0.624	+0.41	-3.14
0.2350	HARPS	+0.420	0.250	+0.578	0.546	+0.85	+0.48
0.2594	HARPS	+0.417	0.252	+0.496	0.554	+0.83	+0.32
0.2636	FEROS	+0.198	0.190	+0.064	0.472	-0.05	-0.54
0.3398	FEROS	+0.131	0.215	+0.140	0.421	-0.36	-0.42
0.3541	FEROS	+0.108	0.216	+0.626	0.474	-0.46	+0.65
0.3748	FEROS	+0.458	0.207	-0.326	0.478	+1.21	-1.35
0.4011	FEROS	+0.023	0.226	+0.169	0.553	-0.82	-0.27
0.5982	FEROS	+0.093	0.220	+0.679	0.483	-0.52	+0.75
0.6198	FEROS	+0.041	0.205	+0.709	0.466	-0.81	+0.84
0.6327	FEROS	+0.082	0.196	+0.718	0.386	-0.64	+1.04
0.6516	FEROS	+0.178	0.229	+1.085	0.513	-0.13	+1.49
0.9012	FEROS	+0.266	0.217	+1.267	0.448	+0.26	+2.12

Table A.2. AK For (FEROS + HARPS).

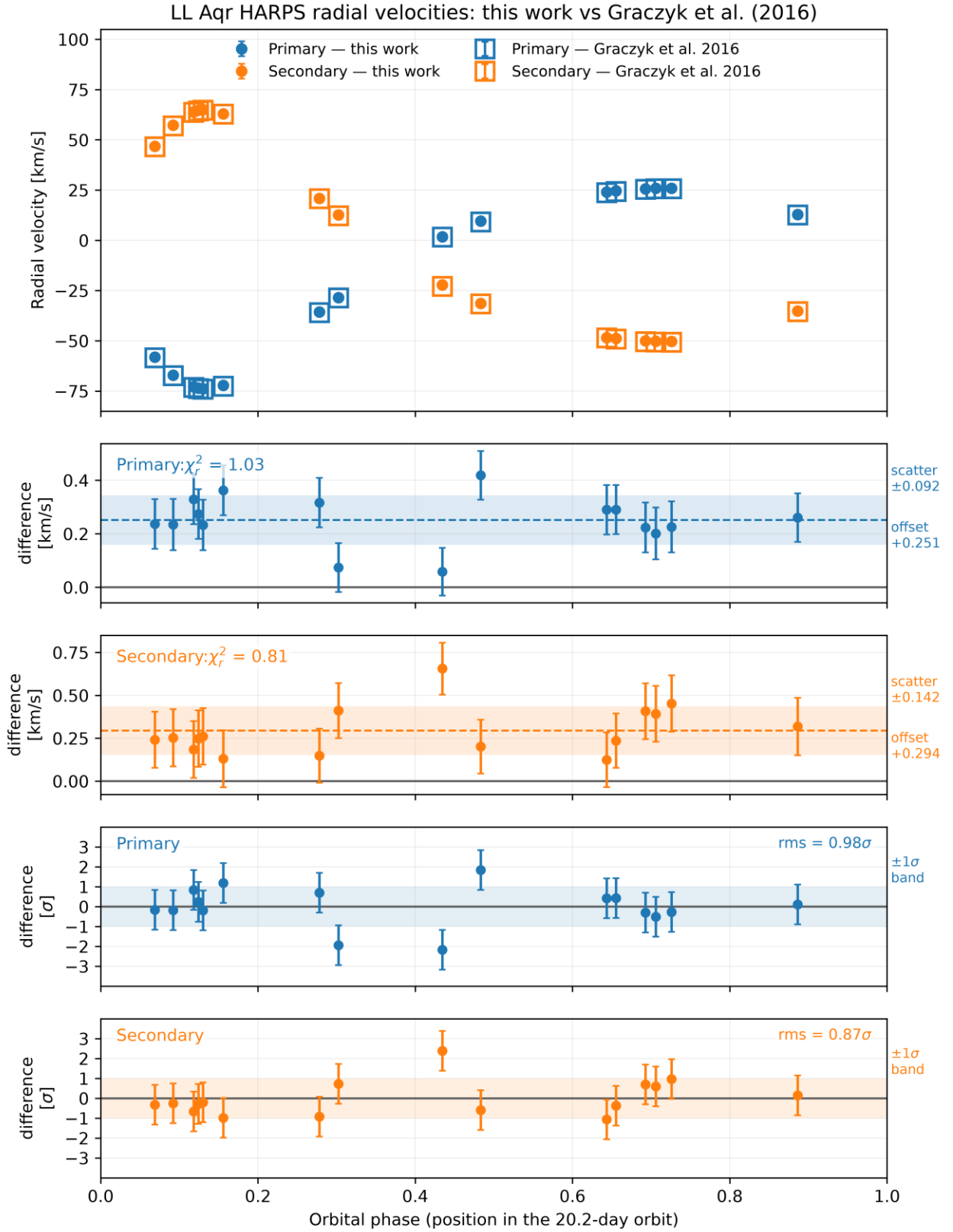


Figure 1. LL Aqr, this work versus Graczyk et al. (2016), for the 16 nights present in both data sets. Top: phased velocity curves (filled circles: this work; open squares: published). Middle: differences per component, with the weighted mean offset (dashed) and its scatter (band). Bottom: the same residuals in units of their own 1σ uncertainty after removing the offset, on a fixed $\pm 3\sigma$ scale.

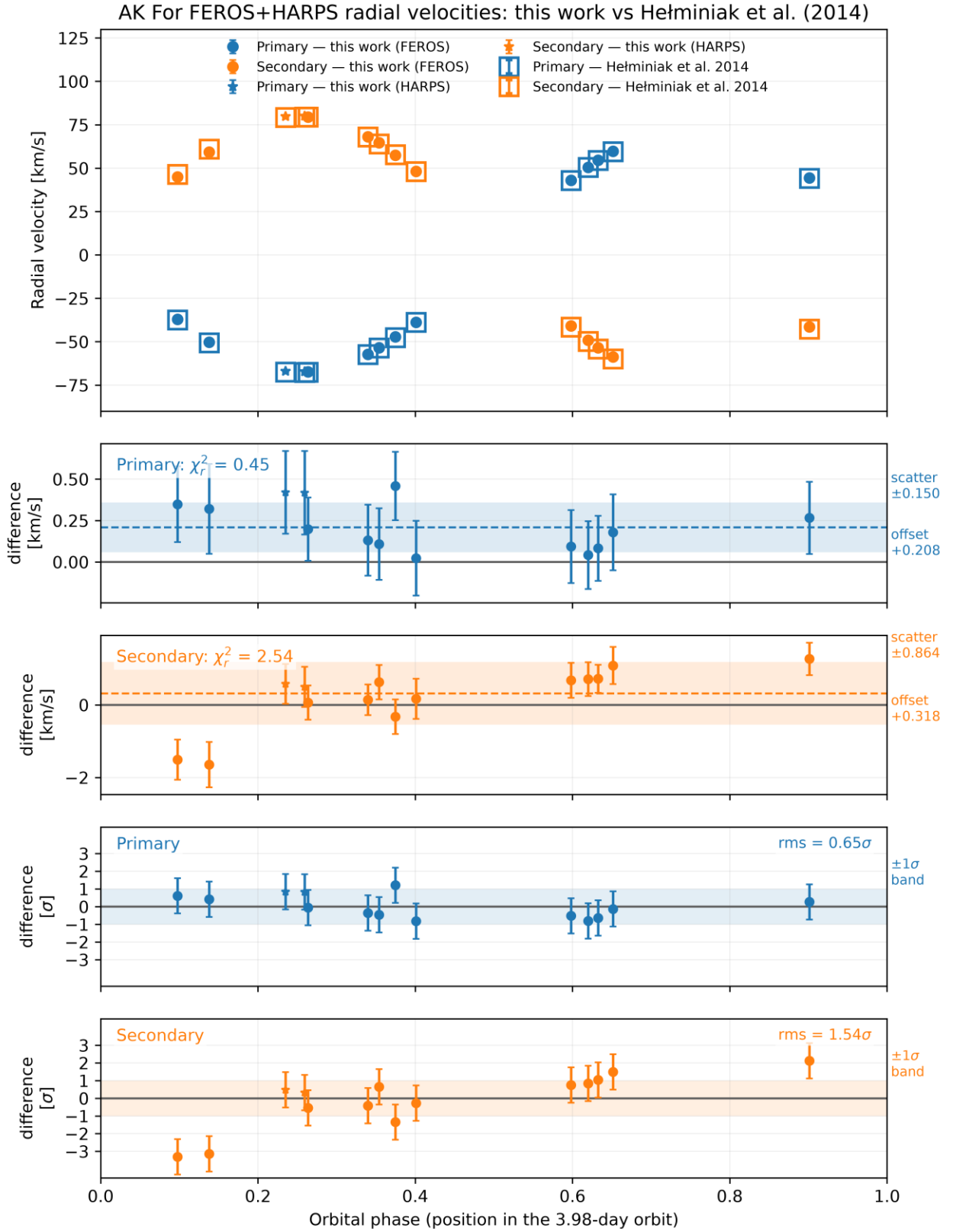


Figure 2. AK For, this work (circles: FEROS; stars: HARPS) versus Hełminiak et al. (2014), for the 14 matched nights; panels as in Figure 1. The primary residuals scatter about a constant offset, while the secondary residuals slope from negative at phase 0.1 to positive beyond phase 0.6 — the signature of a slightly too small velocity amplitude.